

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673



231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

March 24, 2026

Patrick Miller
Cadillac, City of
1121 Plett Rd.
Cadillac, MI 49601

RE: Trace Project 25E1239
Client Project NPDES AMR.- Tert. Effluent 5/22/25

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

A handwritten signature in black ink, appearing to read "Timothy W. Brewer".

Tim Brewer
Project Manager
Enclosures



TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25E1239-01	Tert. Effluent Composite	Aqueous	tl	05/22/25 07:00	05/22/25 13:25
Sample type: Composite					
25E1239-02	Tert. Effluent Grab	Aqueous	tl	05/22/25 09:25	05/22/25 13:25
Sample type: Grab					

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-01 Matrix: Aqueous Date Collected: 05/22/25 07:00
Sample ID: Tert. Effluent Composite Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
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METALS, TOTAL

Analysis Method: EPA 200.7 Rev. 4.4

Batch: T167912

Calcium	76 mg/L	1.0	1	06/02/25	ejb	06/03/25	cma		
Magnesium	18 mg/L	1.0	1	06/02/25	ejb	06/03/25	cma		

Analysis Method: EPA 200.8 Rev. 5.4

Batch: T167912

Antimony	0.42 ug/L	0.25	1	06/02/25	ejb	06/03/25	jma		
Chromium	0.58 ug/L	5.0	1	06/02/25	ejb	06/03/25	jma	J	
Lead	0.14 ug/L	0.55	1	06/02/25	ejb	06/03/25	jma	J	
Nickel	2.7 ug/L	5.0	1	06/02/25	ejb	06/03/25	jma	J	
Selenium	0.60 ug/L	0.50	1	06/02/25	ejb	06/03/25	jma		
Thallium	<1.0 ug/L	1.0	1	06/02/25	ejb	06/03/25	jma		
Zinc	13 ug/L	10	1	06/02/25	ejb	06/03/25	jma		

Analysis Method: SM 2340 B-21

Batch: [CALC]

Hardness as CaCO3	260 mg/L	2.6	1	06/02/25		06/03/25	cma	N	
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SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS

Analysis Method: EPA 625.1

Batch: T167783

N-Nitrosodimethylamine	<4.8 ug/L	4.8	1	05/29/25	kbc	06/02/25	jh		
Bis(2-chloroethyl)ether	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
2-Chlorophenol	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Phenol	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
1,4-Dichlorobenzene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Bis(2-chloroisopropyl)ether	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
N-Nitrosodi-n-propylamine	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Nitrobenzene	0.98 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Isophorone	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
2-Nitrophenol	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
2,4-Dimethylphenol	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Bis(2-chloroethoxy)methane	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-01 Matrix: Aqueous Date Collected: 05/22/25 07:00
Sample ID: Tert. Effluent Composite Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
1,2,4-Trichlorobenzene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
2,4-Dichlorophenol	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Naphthalene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
4-Chloro-3-methylphenol	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
2,4,6-Trichlorophenol	<1.7 ug/L	1.7	1	05/29/25	kbc	06/02/25	jh		
2-Chloronaphthalene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Dimethyl phthalate	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Acenaphthylene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
2,6-Dinitrotoluene	1.3 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh	J	
Acenaphthene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
2,4-Dinitrotoluene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
4-Nitrophenol	<19 ug/L	19	1	05/29/25	kbc	06/02/25	jh		
2,4-Dinitrophenol	<19 ug/L	19	1	05/29/25	kbc	06/02/25	jh		
Diethyl phthalate	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Fluorene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
4-Chlorophenyl phenyl ether	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
4,6-Dinitro-2-methylphenol	<4.8 ug/L	4.8	1	05/29/25	kbc	06/02/25	jh		
N-Nitrosodiphenylamine	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
1,2-Diphenylhydrazine	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh	N	
4-Bromophenyl phenyl ether	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Phenanthrene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Anthracene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Di-n-butyl phthalate	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Fluoranthene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Pyrene	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Butyl benzyl phthalate	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Benzo (a) anthracene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Chrysene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
3,3'-Dichlorobenzidine	<4.8 ug/L	4.8	1	05/29/25	kbc	06/02/25	jh		
Bis(2-ethylhexyl)phthalate	<2.4 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh		
Di-n-octyl phthalate	1.6 ug/L	2.4	1	05/29/25	kbc	06/02/25	jh	J	
Benzo (b) fluoranthene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Benzo (k) fluoranthene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-01 Matrix: Aqueous Date Collected: 05/22/25 07:00
Sample ID: Tert. Effluent Composite Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
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SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS

Benzo (a) pyrene	0.64 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh	J	
Indeno (1,2,3-cd) pyrene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Dibenz (a,h) anthracene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Benzo (g,h,i) perylene	<0.95 ug/L	0.95	1	05/29/25	kbc	06/02/25	jh		
Surrogates:									
2-Fluorophenol	30 %	20-53	1	05/29/25	kbc	06/02/25	jh		
Phenol-d5	18 %	11-40	1	05/29/25	kbc	06/02/25	jh		
Nitrobenzene-d5	82 %	36-103	1	05/29/25	kbc	06/02/25	jh		
2-Fluorobiphenyl	81 %	36-119	1	05/29/25	kbc	06/02/25	jh	N	
2,4,6-Tribromophenol	73 %	30-105	1	05/29/25	kbc	06/02/25	jh	N	
Terphenyl-d14	60 %	37-109	1	05/29/25	kbc	06/02/25	jh		

Analysis Method: EPA 625.1 SIM

Batch: T167626

Benzidine	<0.10 ug/L	0.10	1	05/27/25	als	05/29/25	jh		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	05/27/25	als	05/29/25	jh		
Surrogates:									
Nitrobenzene-d5	102 %	38-130	1	05/27/25	als	05/29/25	jh		
2-Fluorobiphenyl	67 %	39-118	1	05/27/25	als	05/29/25	jh	N	
Terphenyl-d14	62 %	14-96	1	05/27/25	als	05/29/25	jh		

PESTICIDES/PCBS

Analysis Method: EPA 612

Batch: T167557

Hexachlorobenzene	<0.0094 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	N	
Hexachlorobutadiene	<0.0094 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	N	
Hexachlorocyclopentadiene	0.0046 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	J, N	
Hexachloroethane	<0.0094 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	N	
Pentachlorophenol	<1.7 ug/L	1.7	1	05/23/25	kbc	05/23/25	ahr	N	
Surrogates:									
Tetrachloro-m-xylene	61 %	18-105	1	05/23/25	kbc	05/23/25	ahr	N	

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-02 Matrix: Aqueous Date Collected: 05/22/25 09:25
Sample ID: Tert. Effluent Grab Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
<i>Batch: T167559</i>									
Chloromethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Vinyl chloride	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Bromomethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Chloroethane	<2.0 ug/L	2.0	1	05/23/25	kl	05/23/25	kl		
Acrolein	<10 ug/L	10	1	05/23/25	kl	05/23/25	kl		
1,1-Dichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Methylene chloride	<5.0 ug/L	5.0	1	05/23/25	kl	05/23/25	kl		
Acrylonitrile	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
trans-1,2-Dichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1-Dichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Chloroform	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1,1-Trichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Carbon tetrachloride	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Benzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,2-Dichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Trichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,2-Dichloropropane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
cis-1,2-Dichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Bromodichloromethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
2-Chloroethylvinyl ether	<5.0 ug/L	5.0	1	05/23/25	kl	05/23/25	kl		
cis-1,3-Dichloropropene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Toluene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
trans-1,3-Dichloropropene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1,2-Trichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Tetrachloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Dibromochloromethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Chlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Ethylbenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Bromoform	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1,2,2-Tetrachloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,3-Dichlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,4-Dichlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-02 Matrix: Aqueous Date Collected: 05/22/25 09:25
Sample ID: Tert. Effluent Grab Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
1,2-Dichlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Surrogates:									
1,2-Dichloroethane-d4	110 %	68-133	1	05/23/25	kl	05/23/25	kl		
Toluene-d8	92 %	75-120	1	05/23/25	kl	05/23/25	kl		
4-Bromofluorobenzene	101 %	69-119	1	05/23/25	kl	05/23/25	kl		
1,2-Dichlorobenzene-d4	101 %	72-127	1	05/23/25	kl	05/23/25	kl		

WET CHEMISTRY

Analysis Method: EPA 420.4 Rev. 1.0

Batch: T167976

Phenolics	<10 ug/L	10	1	05/30/25	pn	05/30/25	pn		
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QUALITY CONTROL RESULTS

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167912

Analysis Description: Calcium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.7 Rev. 4.4

METHOD BLANK: T167912-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Calcium	mg/L	<1.0	1.0	
Magnesium	mg/L	<1.0	1.0	

LABORATORY CONTROL SAMPLE: T167912-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Calcium	mg/L	16.00	14.6	91	85-115	
Magnesium	mg/L	16.00	15.6	98	85-115	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167912

Analysis Description: Thallium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T167912-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chromium	ug/L	<0.25	0.25	
Nickel	ug/L	<5.0	5.0	
Lead	ug/L	<0.55	0.55	
Antimony	ug/L	<0.25	0.25	
Selenium	ug/L	<0.50	0.50	
Thallium	ug/L	<0.38	0.38	
Zinc	ug/L	<1.2	1.2	

LABORATORY CONTROL SAMPLE: T167912-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chromium	ug/L	50.00	47.4	95	85-115	
Nickel	ug/L	1600	1400	87	85-115	
Lead	ug/L	100.0	101	101	85-115	
Antimony	ug/L	100.0	106	106	85-115	
Selenium	ug/L	100.0	105	105	85-115	

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LABORATORY CONTROL SAMPLE: T167912-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Thallium	ug/L	100.0	102	102	85-115	
Zinc	ug/L	1600	1470	92	85-115	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: [CALC]

Analysis Description: Hardness (Metals)

QC Batch Method:

Analysis Method: SM 2340 B-21

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167557

Analysis Description: 612 Chlorinated Hydrocarbons

QC Batch Method: EPA 3510C Separatory Funnel
Liquid-Liquid Extr.

Analysis Method: EPA 612

METHOD BLANK: T167557-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Hexachlorobenzene	ug/L	0.00760	0.010	J
Hexachlorobutadiene	ug/L	<0.010	0.010	
Hexachlorocyclopentadiene	ug/L	0.00480	0.010	J
Hexachloroethane	ug/L	<0.010	0.010	
Pentachlorophenol	ug/L	<1.8	1.8	
Tetrachloro-m-xylene (S)	%	82	18-105	

LABORATORY CONTROL SAMPLE: T167557-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Hexachlorobenzene	ug/L	0.05000	0.0358	72	26-128	
Hexachlorobutadiene	ug/L	0.05000	0.0242	48	26-107	
Hexachlorocyclopentadiene	ug/L	0.05000	0.0259	52	25-105	
Hexachloroethane	ug/L	0.05000	0.0274	55	26-87	
Pentachlorophenol	ug/L	1.000	<1.8	75	25-100	J
Tetrachloro-m-xylene (S)	%	0.06000	0.0480	80	18-105	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167783

Analysis Description: TTO Semi-Volatiles

QC Batch Method: EPA 625 Extraction for BNAs

Analysis Method: EPA 625.1

CERTIFICATE OF ANALYSIS

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METHOD BLANK: T167783-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
N-Nitrosodimethylamine	ug/L	3.04	5.0	J
Bis(2-chloroethyl)ether	ug/L	<1.0	1.0	
2-Chlorophenol	ug/L	<2.5	2.5	
Phenol	ug/L	<2.5	2.5	
1,4-Dichlorobenzene	ug/L	<2.5	2.5	
Bis(2-chloroisopropyl)ether	ug/L	<2.5	2.5	
N-Nitrosodi-n-propylamine	ug/L	<2.5	2.5	
Nitrobenzene	ug/L	<1.0	1.0	
Isophorone	ug/L	<2.5	2.5	
2-Nitrophenol	ug/L	<2.5	2.5	
2,4-Dimethylphenol	ug/L	<2.5	2.5	
Bis(2-chloroethoxy)methane	ug/L	<2.5	2.5	
1,2,4-Trichlorobenzene	ug/L	<2.5	2.5	
2,4-Dichlorophenol	ug/L	<2.5	2.5	
Naphthalene	ug/L	<2.5	2.5	
4-Chloro-3-methylphenol	ug/L	<2.5	2.5	
2,4,6-Trichlorophenol	ug/L	<1.8	1.8	
2-Chloronaphthalene	ug/L	<2.5	2.5	
Dimethyl phthalate	ug/L	<2.5	2.5	
Acenaphthylene	ug/L	<2.5	2.5	
2,6-Dinitrotoluene	ug/L	<2.5	2.5	
Acenaphthene	ug/L	<2.5	2.5	
2,4-Dinitrotoluene	ug/L	<2.5	2.5	
4-Nitrophenol	ug/L	<20	20	
2,4-Dinitrophenol	ug/L	<20	20	
Diethyl phthalate	ug/L	<2.5	2.5	
Fluorene	ug/L	<2.5	2.5	
4-Chlorophenyl phenyl ether	ug/L	<2.5	2.5	
4,6-Dinitro-2-methylphenol	ug/L	<5.0	5.0	
N-Nitrosodiphenylamine	ug/L	<2.5	2.5	
1,2-Diphenylhydrazine	ug/L	<2.5	2.5	
4-Bromophenyl phenyl ether	ug/L	<2.5	2.5	
Phenanthrene	ug/L	<1.0	1.0	
Anthracene	ug/L	<2.5	2.5	
Di-n-butyl phthalate	ug/L	<2.5	2.5	
Fluoranthene	ug/L	<1.0	1.0	
Pyrene	ug/L	<2.5	2.5	
Butyl benzyl phthalate	ug/L	<2.5	2.5	
Benzo (a) anthracene	ug/L	<1.0	1.0	
Chrysene	ug/L	<1.0	1.0	
Bis(2-ethylhexyl)phthalate	ug/L	<2.5	2.5	
Di-n-octyl phthalate	ug/L	1.71	2.5	J
Benzo (b) fluoranthene	ug/L	<1.0	1.0	
Benzo (k) fluoranthene	ug/L	<1.0	1.0	

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METHOD BLANK: T167783-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzo (a) pyrene	ug/L	0.640	1.0	J
Indeno (1,2,3-cd) pyrene	ug/L	<1.0	1.0	
Dibenz (a,h) anthracene	ug/L	<1.0	1.0	
Benzo (g,h,i) perylene	ug/L	<1.0	1.0	
2-Fluorophenol (S)	%	20	20-53	
Phenol-d5 (S)	%	11	11-40	
Nitrobenzene-d5 (S)	%	56	36-103	
2-Fluorobiphenyl (S)	%	54	36-119	
2,4,6-Tribromophenol (S)	%	56	30-105	
Terphenyl-d14 (S)	%	69	37-109	

LABORATORY CONTROL SAMPLE: T167783-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bis(2-chloroethyl)ether	ug/L	100.0	64.0	64	43-126	
2-Chlorophenol	ug/L	100.0	69.8	70	41-89	
Phenol	ug/L	100.0	19.9	20	17-120	
1,4-Dichlorobenzene	ug/L	100.0	68.5	68	39-78	
Bis(2-chloroisopropyl)ether	ug/L	100.0	55.4	55	63-139	BS06
N-Nitrosodi-n-propylamine	ug/L	100.0	77.0	77	14-198	
Nitrobenzene	ug/L	100.0	76.3	76	54-158	
Isophorone	ug/L	100.0	73.3	73	47-180	
2-Nitrophenol	ug/L	100.0	79.9	80	45-167	
2,4-Dimethylphenol	ug/L	100.0	67.9	68	1-173	
Bis(2-chloroethoxy)methane	ug/L	100.0	76.9	77	49-165	
1,2,4-Trichlorobenzene	ug/L	100.0	81.0	81	57-130	
2,4-Dichlorophenol	ug/L	100.0	77.9	78	54-122	
Naphthalene	ug/L	100.0	80.5	80	36-120	
4-Chloro-3-methylphenol	ug/L	100.0	75.5	76	45-93	
2,4,6-Trichlorophenol	ug/L	100.0	82.2	82	52-129	
2-Chloronaphthalene	ug/L	100.0	89.7	90	65-120	
Dimethyl phthalate	ug/L	100.0	83.2	83	1-120	
Acenaphthylene	ug/L	100.0	89.0	89	54-126	
2,6-Dinitrotoluene	ug/L	100.0	85.1	85	68-137	
Acenaphthene	ug/L	100.0	82.3	82	60-132	
2,4-Dinitrotoluene	ug/L	100.0	84.4	84	48-127	
4-Nitrophenol	ug/L	100.0	20.7	21	13-129	
2,4-Dinitrophenol	ug/L	100.0	73.7	74	1-173	
Diethyl phthalate	ug/L	100.0	83.3	83	1-120	
Fluorene	ug/L	100.0	88.6	89	70-120	
4-Chlorophenyl phenyl ether	ug/L	100.0	82.6	83	38-145	

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LABORATORY CONTROL SAMPLE: T167783-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
4,6-Dinitro-2-methylphenol	ug/L	100.0	83.2	83	53-130	
N-Nitrosodiphenylamine	ug/L	100.0	121	121	50-110	BS06
1,2-Diphenylhydrazine	ug/L	100.0	90.0	90	55-115	
4-Bromophenyl phenyl ether	ug/L	100.0	86.0	86	65-120	
Phenanthrene	ug/L	100.0	90.7	91	65-120	
Anthracene	ug/L	100.0	96.2	96	43-120	
Di-n-butyl phthalate	ug/L	100.0	94.4	94	8-120	
Fluoranthene	ug/L	100.0	104	104	43-121	
Pyrene	ug/L	100.0	98.6	99	70-120	
Butyl benzyl phthalate	ug/L	100.0	109	109	1-140	
Benzo (a) anthracene	ug/L	100.0	95.7	96	43-133	
Chrysene	ug/L	100.0	94.5	94	44-140	
Bis(2-ethylhexyl)phthalate	ug/L	100.0	108	108	29-137	
Di-n-octyl phthalate	ug/L	100.0	99.3	99	19-132	
Benzo (b) fluoranthene	ug/L	100.0	82.6	83	42-140	
Benzo (k) fluoranthene	ug/L	100.0	78.4	78	25-146	
Benzo (a) pyrene	ug/L	100.0	86.3	86	32-148	
Indeno (1,2,3-cd) pyrene	ug/L	100.0	104	104	1-151	
Dibenz (a,h) anthracene	ug/L	100.0	99.6	100	1-200	
Benzo (g,h,i) perylene	ug/L	100.0	118	118	1-195	
2-Fluorophenol (S)	%	100.0	32.4	32	20-53	
Phenol-d5 (S)	%	100.0	19.4	19	11-40	
Nitrobenzene-d5 (S)	%	100.0	87.4	87	36-103	
2-Fluorobiphenyl (S)	%	100.0	83.2	83	36-119	
2,4,6-Tribromophenol (S)	%	100.0	99.1	99	30-105	
Terphenyl-d14 (S)	%	100.0	104	104	37-109	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167626

Analysis Description: 625 Benzidines SIM

QC Batch Method: EPA 625 Extraction for BNAs

Analysis Method: EPA 625.1 SIM

METHOD BLANK: T167626-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	103	38-130	
2-Fluorobiphenyl (S)	%	69	39-118	
Terphenyl-d14 (S)	%	79	14-96	

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LABORATORY CONTROL SAMPLE: T167626-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.000	1.16	58	16-104	
3,3'-Dichlorobenzidine	ug/L	2.000	2.73	136	53-133	BS05
Nitrobenzene-d5 (S)	%	1.000	0.985	99	38-130	
2-Fluorobiphenyl (S)	%	1.000	0.671	67	39-118	
Terphenyl-d14 (S)	%	1.000	0.785	79	14-96	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167559

Analysis Description: TTO Volatiles

QC Batch Method: EPA 624 Purge and Trap

Analysis Method: EPA 624.1

METHOD BLANK: T167559-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chloromethane	ug/L	<1.0	1.0	
Vinyl chloride	ug/L	<1.0	1.0	
Bromomethane	ug/L	<1.0	1.0	
Chloroethane	ug/L	<2.0	2.0	
Acrolein	ug/L	<10	10	
1,1-Dichloroethene	ug/L	<1.0	1.0	
Methylene chloride	ug/L	<5.0	5.0	
Acrylonitrile	ug/L	<2.0	2.0	
trans-1,2-Dichloroethene	ug/L	<1.0	1.0	
1,1-Dichloroethane	ug/L	<1.0	1.0	
Chloroform	ug/L	<1.0	1.0	
1,1,1-Trichloroethane	ug/L	<1.0	1.0	
Carbon tetrachloride	ug/L	<1.0	1.0	
Benzene	ug/L	<1.0	1.0	
1,2-Dichloroethane	ug/L	<1.0	1.0	
Trichloroethene	ug/L	<1.0	1.0	
1,2-Dichloropropane	ug/L	<1.0	1.0	
cis-1,2-Dichloroethene	ug/L	<1.0	1.0	
Bromodichloromethane	ug/L	<1.0	1.0	
2-Chloroethylvinyl ether	ug/L	<5.0	5.0	
cis-1,3-Dichloropropene	ug/L	<1.0	1.0	
Toluene	ug/L	<1.0	1.0	
trans-1,3-Dichloropropene	ug/L	<1.0	1.0	
1,1,2-Trichloroethane	ug/L	<1.0	1.0	
Tetrachloroethene	ug/L	<1.0	1.0	
Dibromochloromethane	ug/L	<1.0	1.0	
Chlorobenzene	ug/L	<1.0	1.0	

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METHOD BLANK: T167559-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Ethylbenzene	ug/L	<1.0	1.0	
Bromoform	ug/L	<1.0	1.0	
1,1,2,2-Tetrachloroethane	ug/L	<1.0	1.0	
1,3-Dichlorobenzene	ug/L	<1.0	1.0	
1,4-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichloroethane-d4 (S)	%	115	68-133	
Toluene-d8 (S)	%	89	75-120	
4-Bromofluorobenzene (S)	%	101	69-119	
1,2-Dichlorobenzene-d4 (S)	%	102	72-127	

LABORATORY CONTROL SAMPLE: T167559-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chloromethane	ug/L	50.00	55.3	111	1-205	
Vinyl chloride	ug/L	50.00	65.6	131	5-195	
Bromomethane	ug/L	50.00	52.8	106	15-185	
Chloroethane	ug/L	50.00	56.8	114	40-160	
Acrolein	ug/L	50.00	40.5	81	60-140	
1,1-Dichloroethene	ug/L	50.00	64.1	128	50-150	
Methylene chloride	ug/L	50.00	59.5	119	60-140	
Acrylonitrile	ug/L	50.00	45.2	90	60-140	
trans-1,2-Dichloroethene	ug/L	50.00	63.4	127	70-130	
1,1-Dichloroethane	ug/L	50.00	62.2	124	70-130	
Chloroform	ug/L	50.00	59.5	119	70-135	
1,1,1-Trichloroethane	ug/L	50.00	64.5	129	70-130	
Carbon tetrachloride	ug/L	50.00	67.6	135	70-130	BS05
Benzene	ug/L	50.00	59.7	119	65-135	
1,2-Dichloroethane	ug/L	50.00	63.0	126	70-130	
Trichloroethene	ug/L	50.00	56.6	113	65-135	
1,2-Dichloropropane	ug/L	50.00	58.6	117	35-165	
cis-1,2-Dichloroethene	ug/L	50.00	63.8	128	70-130	
Bromodichloromethane	ug/L	50.00	64.2	128	65-135	
2-Chloroethylvinyl ether	ug/L	50.00	69.5	139	1-225	
cis-1,3-Dichloropropene	ug/L	50.00	59.8	120	25-175	
Toluene	ug/L	50.00	51.5	103	70-130	
trans-1,3-Dichloropropene	ug/L	50.00	56.6	113	50-150	
1,1,2-Trichloroethane	ug/L	50.00	50.5	101	70-130	
Tetrachloroethene	ug/L	50.00	52.6	105	70-130	
Dibromochloromethane	ug/L	50.00	52.5	105	70-135	
Chlorobenzene	ug/L	50.00	49.8	100	65-135	

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LABORATORY CONTROL SAMPLE: T167559-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Ethylbenzene	ug/L	50.00	51.7	103	60-140	
Bromoform	ug/L	50.00	52.5	105	70-130	
1,1,2,2-Tetrachloroethane	ug/L	50.00	51.0	102	60-140	
1,3-Dichlorobenzene	ug/L	50.00	56.2	112	70-130	
1,4-Dichlorobenzene	ug/L	50.00	54.6	109	65-135	
1,2-Dichlorobenzene	ug/L	50.00	53.1	106	65-135	
1,2-Dichloroethane-d4 (S)	%	30.00	33.3	111	68-133	
Toluene-d8 (S)	%	30.00	27.3	91	75-120	
4-Bromofluorobenzene (S)	%	30.00	30.5	102	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.00	29.5	98	72-127	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167976

Analysis Description: Phenols, Total

QC Batch Method: EPA 420.4 Rev. 1.0

Analysis Method: EPA 420.4 Rev. 1.0

METHOD BLANK: T167976-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenolics	ug/L	<10	10	

LABORATORY CONTROL SAMPLE: T167976-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenolics	ug/L	100.0	108	108	73-110	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

Definitions and Notes

Trace ID: T167559-BS1

Analysis: EPA 624.1

Carbon tetrachloride	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.
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Trace ID: T167626-BS1

Analysis: EPA 625.1 SIM

3,3'-Dichlorobenzidine	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.
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Trace ID: T167626-BSD1

Analysis: EPA 625.1 SIM

3,3'-Dichlorobenzidine	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.
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Trace ID: T167783-BS1

Analysis: EPA 625.1

Bis(2-chloroisopropyl)ether	Note BS06 : Sporadic marginal recovery failure. No data require qualification.
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N-Nitrosodiphenylamine	Note BS06 : Sporadic marginal recovery failure. No data require qualification.
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Trace ID No.

Sampling Time

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25E1239

Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

Date: 5/22/25	Original Observation	Corrected Temperature	IR-9 (CF: +0.1°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: 0.0°C)	SR1 (CF: -0.4°C)	SR2 (CF: -0.1°C)	Temp Blank	Client Sample
Time: 1640									
Initials: MJ									
Package Description: Cooler	Package Temp °C	0.0	0.0						
	Representative Sample Temp °C	3.0	3.0						

Sample Receipt

Yes No

- ☒ Received on ice or other coolant
☒ Ice still present upon receipt
☐ Custody seals present
☒ Trace Courier ☐ Client Drop-off
☐ Yes ☐ No Custody seals intact (if applicable)
☐ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes No N/A

- ☒ All sample containers arrived unbroken and labeled
☒ Sufficient sample to run requested analyses
☒ Correct chemical preservative added to samples
☐ Samples preserved at Trace
☒ Chemical preservation verified, check EMD pH test strip used (if applicable)
☒ pH 0-2.5 (Lot: HC311850) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other
☒ Air bubbles absent from VOAs

Chain of Custody (COC)

Yes No

- ☒ All bottle labels agree with COC
☒ COC filled out properly
☒ COC signed by client

Notes:

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